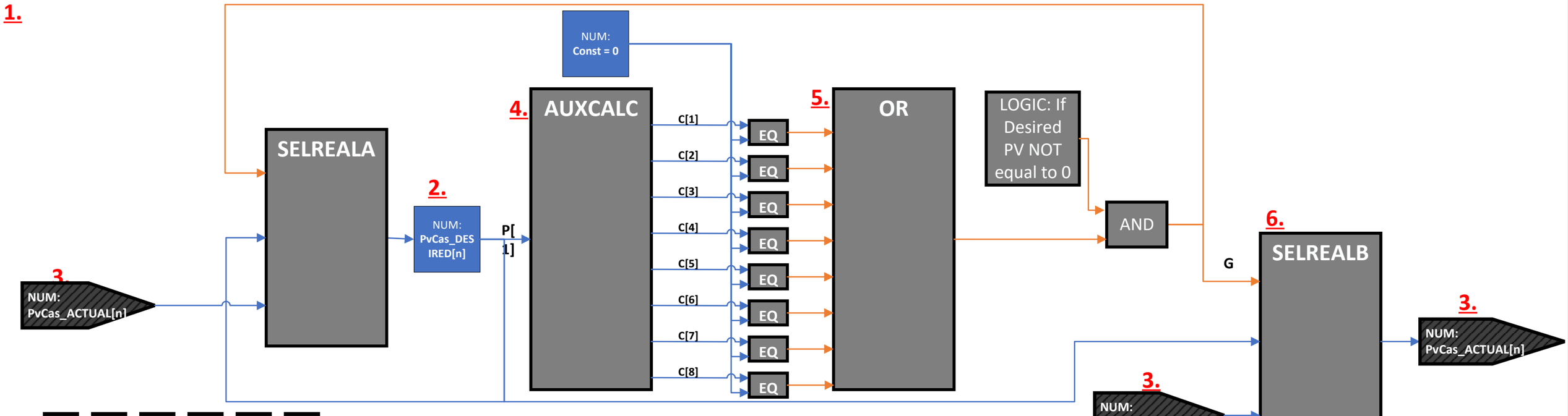
July 29, 2022

PP-0008 PPServer1 PV Routing [Final CM Component Design] REV0



LEGEND

NUMERIC INPUT

ROUTED NUMERIC VARIABLE

PROGRAM BLOCK

AUXCALC Expressions: {LET: PvCas_ACTUAL[n] = X[n]}

C[1] := (X[1] = P[1])? 1 : (X[2] = P[1])? 1 : (X[3] = P[1])? 1 : 0

C[2] := (X[4] = P[1])? 1 : (X[5] = P[1])? 1 : (X[6] = P[1])? 1 : 0

C[3] := (X[7] = P[1])? 1 : (X[8] = P[1])? 1 : (X[9] = P[1])? 1 : 0

C[4] := (X[10] = P[1])? 1 : (X[11] = P[1])? 1 : (X[12] = P[1])? 1 : 0

C[5] := (X[13] = P[1])? 1 : (X[14] = P[1])? 1 : (X[15] = P[1])? 1 : 0

C[6] := (X[16] = P[1])? 1 : (X[17] = P[1])? 1 : (X[18] = P[1])? 1 : 0

C[7] := (X[19] = P[1])? 1 : (X[20] = P[1])? 1 : (X[21] = P[1])? 1 : 0

C[8] := (X[22] = P[1])? 1 : (X[23] = P[1])? 1 : (X[24] = P[1])? 1 : 0

NOTE: Identical process loop for both PV1 and PV2 selection for each controller PV routing selection CM

NOTES:

1) CASCADE PV routing will exist for each assignable MV, thus, each MV will be able to have a inner-loop PV assigned to it.
2) EXECUTION ORDER VITAL TO CM FUNCTION:
   a) SELREALA must be first to run in CM, with PvCas_DESIRED[n] to run immediately after.
   b) Purpose is that NUM can be manipulated at any time from HMI, when NUM is changed by operator SELREAL must run first to inherit new value of PvCas_DESIRED[n]. This way on next iteration of CM:
   IF Desired CASCADE PV is already in use by another cascade loop (i.e., OR output = true), PvCas_ACTUAL[n] is written to PvCas_DESIRED[n], thus desired value is reverts to old variable.
   ELSE PvCas_DESIRED[n] is written to PvCas_ACTUAL[n], thus CASCADE PV becomes new desired variable.
3) Routed from/to PID_PvSp_ARRAY-R1A CM.
4) a) Each existing incident of AUXCALC based Cascade PV routing will be identical except the corresponding PvCas_ACTUAL[n] will be omitted from the appropriate expression (i.e., if the routing is being done for PvCas_ACTUAL[3] then expression 1 will become:
   C[1] := (X[1] = P[1])? 1 : (X[2] = P[1])? 1 : (X[3] = P[1])? 1 : 0
   b) It will be appropriate to have one of the PvCas_DESIRED/ACTUAL array entries equivilent to “No Cascade Selected”, most likely the first entry (i.e., PvCas_DESIRED/ACTUAL[1]) thus, the final program will likely omit PvCas_ACTUAL[1] from AUXCALC expressions.
5) If any of the output expressions from AUXCALC are true, (i.e., if Cascade PV is already in use by a controller)
6) If OR output true PvCas_ACTUAL[n] writes to itself, thus, desired value gets overwritten by the already set actual value.
   else, PvCas_DESIRED[n] is moved to PvCas_ACTUAL[n], thus becoming the new CASCADE PV

Revision History
Revision 0A 18-07-2022
Revision 0B 26-07-2022
Revision 0C 29-07-2022
Revision 0 06-12-2022